SINGAPORE DIMENSIONS • GRADE 6A • CHAPTER 3

# Algebraic Expressions & Postfix Traversal

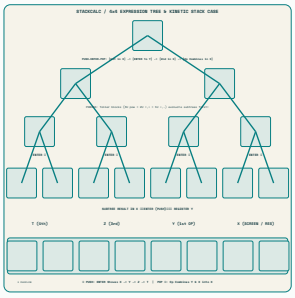
Tactile Concrete-Pictorial-Abstract (CPA) Station Guide • CCSS.MATH.6.EE.A.2

Student Name: _____ Date: _____ **Model: SC-6** **Check Score: ___ / 4**

**STUDENT SELF-CHECK RULE (NO CHEATING):**
Build with blocks and solve on paper first! Use your SC-6 calculator ONLY to verify answers on the 4-line LCD screen.

## STATION LAB TUTORIAL: BENCH SCHEMATIC & CPA PROGRESSION

### 1. 2D Bench Blueprint



Kit: Tree Board

### 2. Desk Setup (Concrete -> Pictorial)

**Physical Bench Setup:**
Build cube towers for variables a=6, b=2, c=8, d=4.

**Sketch Model on Paper:**
Draw binary expression syntax tree with operator root.

### 3. Step-by-Step LCD Stack Trace (SC-6)

MODE: 123 Base-10 Arithmetic

| | | | |
|---|---|---|---|
| 1. | 1 | Step 1 | Execute [1] |
| 2. | 5 | Step 2 | Execute [5] |
| 3. | enter | Step 3 | Execute [enter] |
| 4. | 5 | Step 4 | Execute [5] |

EXPECTED LCD X: 100

**1** **Evaluate (12 - 4) × (3 + 2).**

Manipulative Model & Desk Sketch (Show your work):

**Final Math Answer:** _____

☐ Verified on SC-6

**2** **Evaluate (25 + 15) ÷ (10 - 2).**

Manipulative Model & Desk Sketch (Show your work):

**Final Math Answer:** _____

☐ Verified on SC-6

**3** **Evaluate 8 × (4 + 6) - 20.**

Manipulative Model & Desk Sketch (Show your work):

**Final Math Answer:** _____

☐ Verified on SC-6

**4** **Evaluate (50 - 10) ÷ (2 × 4).**

Manipulative Model & Desk Sketch (Show your work):

**Final Math Answer:** _____

☐ Verified on SC-6

# Station Diagnostic & Solution Key: Chapter 3

Algebraic Expressions & Postfix Traversal • Manipulative Setup, Diagnostic Rubric & Verification Flow

## MANIPULATIVE HARDWARE STATION INVENTORY & 3D PRINT PREPARATION

**Station Kit: Interlocking Cubes & 4-Line Hardware LCD Stack**

CAD Blueprint: tree_board

**Target Device: SC-6 Elementary / SC-32 Bridge**

**CAD / STL Source File:**

Education/build/scad/tree_board.scad

Standard: Gridfinity 42mm Pitch / 0.6mm Inset Markings

**Recommended Filaments:**

Charcoal 4x4 Tree Case & Teal Operation Blocks (PLA)

| # | Math Problem Prompt | Expected Math Answer | Self-Check Keystrokes & RPN Stack |
|---|---|---|---|
| 1 | **Evaluate (12 - 4) × (3 + 2).**<br>Standard: CCSS.MATH.6.EE.A.2 | 40 | [12] [ENTER] [4] [-] [3] [ENTER] [2] [ |
| 2 | **Evaluate (25 + 15) ÷ (10 - 2).**<br>Standard: CCSS.MATH.6.EE.A.2 | 5 | [25] [ENTER] [15] [+] [10] [ENTER] [2] |
| 3 | **Evaluate 8 × (4 + 6) - 20.**<br>Standard: CCSS.MATH.6.EE.A.2 | 60 | [4] [ENTER] [6] [+] [8] [×] [20] [-] |
| 4 | **Evaluate (50 - 10) ÷ (2 × 4).**<br>Standard: CCSS.MATH.6.EE.A.2 | 5 | [50] [ENTER] [10] [-] [2] [ENTER] [4] |

## TEACHER DIAGNOSTIC GUIDE: COMMON MISCONCEPTIONS & REMEDIATION

**Key Misconception to Target During Station Observation:**

Highlight that RPN evaluates expressions strictly from the inside out.

**How the Manipulatives & Calculator Foster Self-Correction:**

Students solve with pencil and physical blocks first. When they self-check on the SC-6, any discrepancy between their written answer and the calculator's 4-line LCD immediately prompts them to re-inspect their physical bench model or regrouping, turning errors into productive struggle without teacher intervention.

**Station Facilitation & Pacing Tip:**

Keep 3D printed trays docked on standard Gridfinity baseplates to prevent loose parts on student desks. Encourage students to articulate their CPA steps: name the block representation (Concrete), point to their drawn tape/bond model (Pictorial), and verify register values (Abstract).

## CLASSROOM SCORING & STATION MASTERY RUBRIC

| | |
|---|---|
| **Exemplary (18–20 pts):** | Complete, accurate manipulative sketches; exact math answers; independently verified self-check on LCD stack. |
| **Proficient (14–17 pts):** | Correct math answers; minor sketch omissions; verified on calculator with minimal teacher prompting. |
| **Developing (10–13 pts):** | Understands problem concept but made computational slips; successfully self-corrected using calculator LCD. |
| **Intervention (<10 pts):** | Relied on calculator before solving on paper, or struggled with underlying physical manipulative models. |